Supplemental Material of "Entropy-Guided Semantic Event-Driven Control for O-RAN": Definitions and Proofs of Propositions

Somreeta Pramanik, Abhinandan Mandal, Soumyapriya Goswami, Samriddha Sanyal, Deborsi Basu, Chiara Lombardo, Roberto Bruschi, Pushpita Chatterjee, and Uttam Ghosh

Appendix A
Event-Driven Semi-Markov Formulation

Model. Let t index E2 reports and $t_0 < t_1 < \dots$ the event epochs, i.e., the reports with e_t=1 in (**??**), with holding times $\tau_k = t_{k+1} - t_k \in \{1, 2, \dots\}$. At epoch k the agent observes s_k, selects $a_k \in \mathcal{A}$, and holds it until the next event. We assume that the event-time process is Markov: given (s_k, a_k), the joint law of (s_{k+1}, τ_k) and of the rewards in the interval is independent of earlier history. With per-report reward r_t, $|r_t| \le R_{\max}$, and discount $\gamma \in (0, 1)$ per report, the discounted reward of interval k is

$$R_k = \sum_{j=0}^{\tau_k - 1} \gamma^j \, r_{t_k + j}, \qquad |R_k| \le \frac{R_{\max}}{1-\gamma}, \tag{1}$$

and the continuation value is discounted by γ^{τ_k}, since τ_k reports have elapsed. The Bellman optimality operator is

$$(\mathcal{T}Q)(s,a) = \mathbb{E}\Big[R_k + \gamma^{\tau_k} \max_{a'} Q(s_{k+1}, a') \,\Big|\, s_k{=}s, a_k{=}a \Big]. \tag{2}$$

For a bounded potential Φ, $|\Phi(s)| \le M$, the shaped interval reward is $R'_k = R_k + F_k$ with the duration-aware term

$$F_k = \gamma^{\tau_k} \Phi(s_{k+1}) - \Phi(s_k). \tag{3}$$

Theorem 1. *$\mathcal{T}$ is a contraction in the supremum norm with modulus at most γ and has a unique bounded fixed point Q^*. Replacing R_k with R'_k preserves the set of optimal policies.*

Proof: Contraction. Let Q_1, Q_2 be bounded. In $(\mathcal{T}Q_1 - \mathcal{T}Q_2)(s,a)$ the rewards cancel, and since $|\max_{a'} x_{a'} - \max_{a'} y_{a'}| \le \max_{a'} |x_{a'} - y_{a'}|$,

$$\begin{aligned} |(\mathcal{T}Q_1 - \mathcal{T}Q_2)(s,a)| &\le \mathbb{E}\big[\gamma^{\tau_k} \,\big|\, s, a\big] \, \|Q_1 - Q_2\|_\infty \\ &\le \gamma \, \|Q_1 - Q_2\|_\infty, \end{aligned}$$

because $\tau_k \ge 1$ implies $\gamma^{\tau_k} \le \gamma$. Taking the supremum over (s,a) gives $\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_\infty \le \gamma \|Q_1 - Q_2\|_\infty$, and the Banach fixed-point theorem yields a unique bounded fixed point Q^*.

Telescoping. Since $t_{k+1} = t_k + \tau_k$, we have $\gamma^{t_k - t_0} \gamma^{\tau_k} = \gamma^{t_{k+1} - t_0}$, so the discounted shaping terms over the first N intervals are

$$\begin{aligned} \sum_{k=0}^{N-1} \gamma^{t_k - t_0} F_k &= \sum_{k=0}^{N-1} \Big(\gamma^{t_{k+1} - t_0} \Phi(s_{k+1}) - \gamma^{t_k - t_0} \Phi(s_k) \Big) \\ &= \gamma^{t_N - t_0} \Phi(s_N) - \Phi(s_0). \end{aligned}$$

As $t_N - t_0 \ge N$, the first term is bounded by $\gamma^N M \to 0$.

Policy invariance. Hence, for any policy π, the shaped and original discounted returns satisfy $G'^{\pi} = G^{\pi} - \Phi(s_0)$, so $Q'^{\pi}(s,a) = Q^{\pi}(s,a) - \Phi(s)$. The shift $-\Phi(s)$ does not depend on a, so $\arg\max_a Q'^{\pi}(s,a) = \arg\max_a Q^{\pi}(s,a)$ in every state, and the set of optimal policies is unchanged [1]. □

Using the per-report term $\gamma\Phi(s_{k+1}) - \Phi(s_k)$ instead of (3) (refer to equation 3 in the main paper) is not equivalent when holding times vary, since the intermediate potentials then no longer cancel.

Appendix B
Boundedness of the Confidence

Lemma 1. *Let $p_h = (p_{h,1}, \dots, p_{h,K_h})$ be a probability vector, $p_{h,k} \ge 0$, $\sum_k p_{h,k} = 1$, with $K_h \ge 2$, and let $H_h = -\sum_k p_{h,k} \log p_{h,k}$ (with $0 \log 0 = 0$). Then $0 \le H_h \le \log K_h$, and the confidence $c_h = 1 - H_h / \log K_h$ in Equation 3 (refer to in the main paper) satisfies $c_h \in [0,1]$, with $c_h = 1$ if and only if p_h is one-hot and $c_h = 0$ if and only if p_h is uniform.*

Proof: Lower bound. Since $0 \le p_{h,k} \le 1$, $\log p_{h,k} \le 0$ and each term $-p_{h,k} \log p_{h,k} \ge 0$; hence $H_h \ge 0$, with equality if and only if one class has probability one.

Upper bound. Writing $H_h = \sum_{k: p_{h,k} > 0} p_{h,k} \log(1/p_{h,k})$ and applying Jensen's inequality to the concave logarithm,

$$\begin{aligned} H_h &\le \log\Big(\sum_{k: p_{h,k} > 0} p_{h,k} \, \frac{1}{p_{h,k}} \Big) \\ &= \log \big| \{ k : p_{h,k} > 0 \} \big| \le \log K_h, \end{aligned}$$

with equality if and only if $p_{h,k} = 1/K_h$ for all k.

Bounds on c_h. Since $K_h \ge 2$, $\log K_h > 0$, and dividing $0 \le H_h \le \log K_h$ by $\log K_h$ gives $0 \le H_h / \log K_h \le 1$. Subtracting from one reverses the inequalities, so $0 \le c_h \le 1$, with the equality cases inherited from the two bounds. □

S. Pramanik, R. Bruschi, C. Lombardo are with S2N lab, CNIT and University of Genova, Italy. A.Mandal is with IIITM Gwalior, India. S.Goswami is with KGEC, India. S.Sanyal is with MathWorks, India. D.Basu is with Ericsson, India. U.Ghosh & P. Chatterjee are with SACS, Meharry Medical College, USA

For a head with a single class ($K_h = 1$), $\log K_h = 0$ and we set $c_h = 1$.

REFERENCES

[1] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Proc. ICML*, 1999, pp. 278–287.